INCH-POUND

MIL-DTL-32434
23 January 2013

DETAIL SPECIFICATION

HOSE ASSEMBLY, POLYTETRAFLUOROETHYLENE, CONVOLUTED, HIGH TEMPERATURE, MEDIUM PRESSURE, GENERAL SPECIFICATION FOR

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains the requirements for medium pressure, high temperature, convoluted polytetrafluoroethylene (PTFE) hose assemblies, reinforced with a corrosion-resistant steel (CRES) wire braid for use in fuel, lubricating oil, water-alcohol, hydraulics and pneumatic systems within the limits specified herein.

1.2 Classification. The hose assemblies are classified as follows (see 6.1):

a. Class 1 - ALL CRES fittings for all sizes, operating temperature ranging from -65°F to +450°F.
b. Class 2 - Combination of aluminum alloy and CRES fittings for sizes 8 and larger, operating temperature ranging from -65°F to +275°F.

1.3 Application limitations.

1.3.1 Size -3/-4 reducer assembly. The size -3/-4 reducer assembly should not be used in aircraft hydraulic systems.

1.3.2 Pneumatic storage system. Pneumatic storage system application should not be used (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, Attn: VAI, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to FluidFlow@dla.mil. Since contact information can change you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-DTL-32435	-	Hose, Polytetrafluoroethylene, Convoluted, 450°, Medium Pressure
MIL-DTL-27272	-	Fittings, Polytetrafluoroethylene Hose, High Temperature, Medium Pressure, General Specification for
MIL-DTL-25579/1	-	Hose Assembly Length Measurement for NAS 1760 and MS8000 Series Seal, (Globe Seal, or Radius Seal) to Flareless Fitting End Design.
MIL-PRF-680	-	Degreasing Solvent
MIL-PRF-5606	-	Hydraulic Fluids, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-PRF-7808	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-87257	-	Hydraulic Fluid, Fire Resistant; Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

DEPARTMENT OF DEFENSE STANDARDS

MS33786	-	Fitting Installation, Flared Tube and Hose, Swivel (ASG)

(See supplement 1 for list of specification sheets.)

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS1760	-	Fitting End, Flareless Acorn, Standard Dimensions for

(Copies of this document is available online at http://www.aia-aerospace.org/ or from the Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3901)

ASTM INTERNATIONAL

ASTM A262	-	Steels, Detecting Susceptibility to Intergranular Attack in Austenitic Stainless
ASTM A276	-	Standard Specification for Stainless Steel Bars and Shapes
ASTM A493	-	Standard Specification for Stainless Steel Wire and Wire Rods for Cold Heading and Cold Forging
ASTM A313/A313M	-	Standard Specification for Stainless Steel Spring Wire
ASTM A580/A580M	-	Steel, Wire, Stainless

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ASTM B210 - Standard Specification for Aluminum and Aluminum-Alloy Drawn Seamless Tubes
ASTM D471 - Rubber Property - Effect of Liquids

(Copies of these documents are available online at http://www.astm.org or from the ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 17025 - General requirements for the competence of testing and calibration laboratories

(Copies of these documents are available online at http://www.ansi.org or from the ANSI Customer Service Department, 25 W. 43rd Street, 4th Floor, New York, NY 10036.)

NCSL INTERNATIONAL

NCSL Z540.3 - Requirements for the Calibration of Measuring and Test Equipment

(Copies of this document is available online at http://www.ncsli.org or from NCSL International 2995 Wilderness Place, Suite 107 Boulder, Colorado 80301-5404)

SAE INTERNATIONAL

SAE-AMS-QQ-A-200/8 - Aluminum Alloy 6061, Bar, Rod, Shapes, Tube, and Wire, Extruded
SAE-AMS-QQ-A-225/4 - Aluminum Alloy, 2014, Bar, Rod, Wire, and Special Shapes; Rolled Drawn, or Cold Finished
SAE-AMS-QQ-A-225/6 - Aluminum Alloy, 2024, Bar, Rod, and Wire; Rolled, Drawn or Cold Finished
SAE-AMS-WW-T-700/4 - Tube, Aluminum Alloy, Drawn, Seamless, 5052
SAE-AMS-WW-T-700/6 - Tube, Aluminum Alloy, Drawn, Seamless, 6061
SAE-AMS-A-22771 - Aluminum Alloy Forgings, Heat Treated
SAE-AMS2700 - Passivation Treatment for Corrosion-Resistant Steel
SAE-AMS4082 - Aluminum Alloy, Seamless Drawn Tubing 1.0Mg - 0.60Si - 0.28Cu - 0.20Cr (6061-T6) Solution and Precipitation Heat Treated - UNS A96061
SAE-AMS4115 - Aluminum Alloy, Rolled or Cold-Finished, Bars, Rods, Wire, and Flash Welded Rings Annealed 1.0Mg - 0.60Si - 0.28Cu - 0.20Cr (6061-0)
SAE-AMS4117 - Aluminum Alloy, Rolled or Cold Finished Bars, Rods, and Wire and Flash Welded Rings 1.0Mg - 0.60Si - 0.28Cu - 0.20Cr (6061; -T6, -T651) Solution and Precipitation Heat Treated - UNS A96061
SAE-AMS4120 - Aluminum Alloy, Rolled or Cold Finished Bars, Rods, and Wire 4.4Cu - 1.5Mg - 0.60Mn (2024) Solution Heat Treated and Naturally Aged (T4) Solution Heat Treated, Cold Worked, and Naturally Aged (T351) - UNS A92024
SAE-AMS4127 - Aluminum Alloy Forgings and Rolled or Forged Rings 1.0 Mg - 0.6 Si - 0.28 Cu - 0.20 Cr (6061-T6), Solution and Precipitation Heat Treated
SAE-AMS5557 - Steel Tubing, Seamless and Welded, Corrosion and Heat Resistant, 18.5 Cr - 10.5 Ni - 0.40 Ti (SAE 30321) Hydraulic, Solution Heat Treated

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SAE-AMS5565	-	Steel, Corrosion Resistant, Welded Tubing 19Cr - 9.5Ni (SAE 30304) Solution Heat Treated
SAE-AMS5566	-	Steel Tubing, Seamless or Welded, Corrosion Resistant, 19 Cr - 10 Ni (SAE 30304) High Pressure Hydraulic, Cold Drawn
SAE-AMS5567	-	Steel Tubing, Seamless or Welded, Corrosion Resistant 19 Cr - 10 Ni (SAE 30304) Hydraulic, Annealed
SAE-AMS5639	-	Steel, Corrosion Resistant, Bars, Wire, Forgings, Tubing, and Rings 19Cr - 9.5NI, Solution Heat Treated
SAE-AMS5643	-	Steel Bars, Forgings, Tubing and Rings, Corrosion Resistant 16.5 Cr - 4.0 Ni - 0.30(Cb + Ta) - 4.0 Cu.
SAE-AMS5645	-	Steel, Corrosion and Heat Resistant, Bars, Wire, Forgings, Tubing, and Rings 18Cr - 10Ni - 0.40Ti (SAE 30321)
SAE-AMS5647	-	Steel, Corrosion Resistant, Bars, Wire, Forgings, Tubing, and Rings 19Cr - 10NI, Solution Heat Treated
SAE-AMS5689	-	Steel Wire, Corrosion and Heat Resistant, 18 Cr - 9.5 Ni - 0.40 Ti (SAE 30321) Solution Heat Treated
SAE-AMS5697	-	Steel, Corrosion Resistant, Wire19Cr - 9.5NI, Solution Heat Treated
SAE-AMS5743	-	Steel, Corrosion and Heat Resistant, Bars and Forgings 15.5 Cr - 4.5 Ni - 2.9 Mo - 0.10 Ni Solution Heat Treated, Sub-Zero Cooled, Equalized and Over-Tempered.
SAE-ARP908	-	Torque Requirements, Installation and Qualification Test, Hose and Tube Fittings
SAE-AS611	-	Polytetrafluoroethylene Hose Assembly, Cleaning Methods
SAE-AS1055	-	Fire Testing of Flexible Hose, Tube Assemblies, Coils, Fittings, and Similar System Components
SAE-AS1946	-	Hose Assembly, Polytetrafluoroethylene, Metallic Reinforced, Up to 1500 psi and 450 Degrees F, Hydraulic and Pneumatic
SAE-AS2078	-	Test Methods, Hose Assemblies, Polytetrafluoroethylene (PTFE)
SAE-AS4375	-	Fitting End, Flareless, Design Standard
SAE-AS4395	-	Fitting End-Flared Tube Connection, Design Standard.
SAE-AS5685	-	Steel, Corrosion Resistant, Safety Wire, 18Cr - 11.5Ni (UNS S30500) Solution Heat Treated, Cold Finished
SAE-AS8879	-	Screw Threads - UNJ Profile, Inch
SAE-AS33514	-	Fitting End, Standard Dimensions for Flareless Tube Connection and Gasket Seal

(Copies of these documents are available online at www.sae.org from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 [inside USA and Canada] or 724-776-4970 [outside USA], email at CustomerService@sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and also in accordance with the applicable specification sheets. In the event of a conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.5.

3.3 Materials. Materials used shall be as specified herein. Materials used that are not specifically specified herein shall be of a quality that will enable the hose assembly to meet the requirements specified herein.

3.3.1 Recycled, recovered, and environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.2 Metals. All metals used herein shall conform to the requirements in table I. Metals used for end fittings (see 3.5.2) shall be corrosion-resistant steel or an aluminum alloy that has been treated to resist corrosion due to fuels, salt spray, and atmospheric conditions encountered by the hose assembly during storage and normal service use.

TABLE I. Requirements for metals.

Metal	Form	Specification	Type
Aluminum alloy	Bars	SAE-AMS-QQ-A-200/8	6061-T6 6061-T651
		SAE-AMS4117	
	Forgings	SAE-AMS4127	6061-T6
		SAE-AMS-A-22771	7075-T73
	Seamless tubing	SAE-AMS-WW-T-700/4	5052, type I
		ASTM B210	6061-T6
		SAE-AMS4082	
		SAE-AMS-WW-T-700/6	
	Wire	SAE-AMS4115 or 4117	6061
		SAE-AMS-QQ-A-225/4	2014
		SAE-AMS-QQ-A-225/6 or SAE-AMS4120	2024-T4 or 2024-T351
Corrosion-resistant steel	Bars & forgings	ASTM A276	Class 304, Condition A or B
			Class 304L, condition A
			Class 321, condition A
		SAE-AMS5639	SAE 30304; UNS S30400
		SAE-AMS5645	SAE 30321; UNS S32100
		SAE-AMS5647	304L; UNS S30403
		SAE-AMS5643	17-4PH
		SAE-AMS5743	AM-355
	Tubing	SAE-AMS5566	Type 1 or 2
		SAE-AMS5557	Type 1 or 2
		SAE-AMS5565	SAE 30304
		SAE-AMS5567	SAE 30304
	Wire	SAE-AMS5689	SAE 30321; UNS S32100
		SAE-AMS5697	SAE 30304; UNS S30400
		ASTM A313/A313M	Type 302, 304, 305, or 321
		ASTM A493 or SAE-AS5685	Type 305, condition A
		ASTM A580/A580M	Type 304, condition A

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3.4 Finish. The following requirements are applicable to permanent and field attachable fittings.

3.4.1 Aluminum parts. Unless otherwise specified, aluminum parts shall be finished in accordance with MIL-A-8625, type II, class 2. Flareless parts shall be dyed yellow and flared parts shall be dyed blue.

3.4.2 Corrosion-resistant steel parts. Unless otherwise specified, corrosion-resistant steel parts shall be passivated in accordance with SAE-AMS2700. Following passivation, all parts shall be thoroughly rinsed in water and dried.

3.5 Design and construction. The hose assembly shall consist of a seamless convoluted polytetrafluoroethylene inner tube, reinforced with corrosion resistant steel reinforcement wires and coupled with corrosion resistant (stainless steel) end fittings (class 1) or aluminum alloy and corrosion resistant end fittings (class 2) suitable for the intended installation. If assemblies with field-attachable fittings are used, the fittings shall comply with MIL-DTL-27272, the hose shall comply with MIL-DTL-32435 and the assemblies shall comply with the requirements of this specification and MIL-DTL-32434/1 through MIL-DTL-32434/5, (see supplement 1) as applicable. Assemblies with permanent fittings shall conform to this specification and MIL-DTL-32434/6 through MIL-DTL-32434/9, as applicable. Standard hose assemblies shall have flared fittings to mate with SAE-AS4395 and flareless fittings shall be in accordance with MIL-DTL-25579/1 for NAS1760, or globe or radius seal design, to mate with SAE-AS33514 or SAE-AS4375 fitting end for flareless tube connections. Hose assembly lengths shall be determined in accordance with MIL-DTL-25579/1. The hose assemblies shall withstand the strain and vibrations encountered during shipment, storage, installation, and service, within the limits specified herein.

3.5.1 Inner tube. The inner tube shall be seamless extrusion of virgin polytetrafluoroethylene resin. The hose inner tube shall conform to all of the requirements in MIL-DTL-32435 for the bulk hose to be assembled with MIL-DTL-27272 hose connector fitting. For permanently attached fitting hose assemblies the hose should meet the requirements of MIL-DTL-32435.

3.5.1.1 Reinforcement. The reinforcement shall consist of corrosion resistant steel wires in accordance with the applicable specification listed in table I Hose under -16Z shall have a single layer of braid and hose -16Z and above shall have two layers of braid. The letter “Z” following the dash size signifies that a double layer of wire braid is mandatory. The wires shall be so arranged over the inner tube as to provide sufficient strength to insure conformance with the requirements herein. Broken or missing reinforcing wires shall be cause for rejection. Crossed-over wires shall not be cause for rejection.

3.5.2 Fittings. The end fittings for the hose assembly may be either field-attachable type in accordance with MIL-DTL-27272 or the permanent type.

3.5.2.1 Materials. Materials shall be selected for the specific operating conditions as specified in 3.3.

3.5.2.2 Class 2 hose assemblies. For class 2 hose assemblies (size -8 and larger), the socket, and the sleeve if used, shall be stainless steel and the nipple, nut, and elbow (if applicable) shall be aluminum alloy, see table I for fitting materials.

3.5.2.3 Flared and flareless fittings. Flared fittings shall mate with SAE-AS4395 and flareless fittings shall mate with SAE-AS33514 or SAE-AS4375, as specified on the individual slash sheets, fitting end for flareless tube connections. Hose assembly lengths shall be determined in accordance with in accordance with the applicable slash sheet in accordance with MIL-DTL-25579/1.

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3.5.2.4 Mating requirements. Flange fittings shall mate with the mounting pad as shown on MS33786. Flared fittings to shall mate with SAE-AS4395 and flareless fittings in accordance with MIL-DTL-25579/1 for NAS1760, or globe or radius seal design to mate with SAE-AS33514 or SAE-AS4375, as specified on the individual slash sheets, fitting end for flareless tube connections.

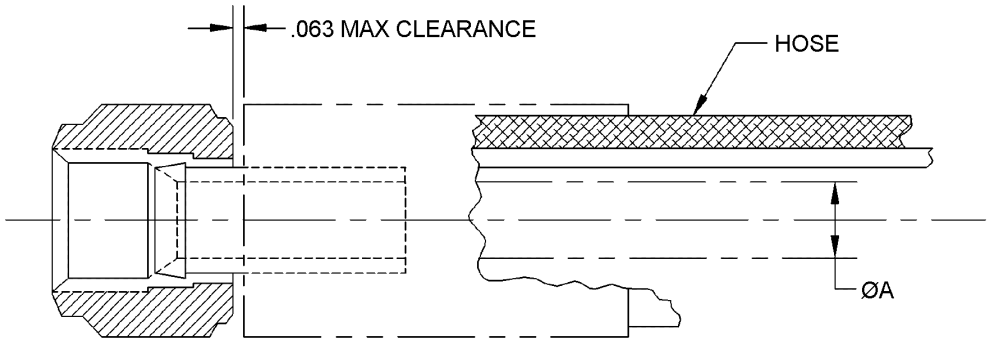
3.5.2.5 Fittings embrittlement resistance. All crimped or swaged end fitting sockets (collars) that are fabricated from 304 stainless steel shall be capable of passing the embrittlement test specified in ASTM A262, practice E, prior to assembly to the nipple or the swaging operation. Sockets fabricated from stabilized austenitic steel are acceptable without being subjected to the embrittlement test.

3.5.3 Screw threads. All coupling nut threads shall be in accordance with SAE-AS8879, category I. However, no more than a 10% increase in the thread tolerances specified in SAE-AS8879 shall be accepted for the coupling nut thread after the first connection (see 4.7.2).

3.5.4 Dimensions and weights. Except for length (see 3.5.5), the dimensions of the hose assembly and the weight of its components shall be as specified in table II and figure 1.

3.5.5 Length. Hose assembly shall be furnished in lengths as specified in the contract or purchase order (see 6.2), however, tolerances on the length of each hose assembly shall be as follows:

a. ±.125 inch for lengths under 18 inches.
b. ±.250 inch for lengths from 18 inches to 36 inches.
c. ±.500 inch for lengths from 36 inches to 50 inches.
d. ±1% for lengths over 50 inches.



Inch	mm
.063	1.60

A dia. (see note 3) Minimum nipple diameter										
Size	-3/-4	-4	-5	-6	-8	-10	-12	-16Z	-20Z	-24Z
inches	.132	.132	.193	.256	.340	.430	.548	.778	1.000	1.250
(mm)	(3.35)	(3.35)	(4.90)	(6.50)	(8.64)	(10.92)	(13.92)	(19.76)	(25.40)	(31.75)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Dimension A is the minimum dimension that may be reduced after hose to fitting assembly.
4. The letter “Z” is used to indicate that two layers of reinforcement are required for these sizes (see MIL-DTL-32435).

FIGURE 1. Hose assembly.

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TABLE II. Dimensions, weights, and ball sizes. 1/ 2/ 3/

Size	Reference OD tubing inches (mm)	Maximum hose weight (lb/ft)	Class 1 maximum fitting weight lbs (kg)	Class 2 maximum fitting weight lbs (kg)	Maximum hose OD inches (mm)	Ball diameter Straight to straight fittings inches (mm) 4/	Ball diameter at least one elbow inches (mm) 4/
-3/-4 5/	.188 (4.78)	.087	.069 (0.031)	--- 6/	.343 (8.71)	.072 (1.83)	.068 (1.73)
-4	.250 (6.35)	.087	.094 (0.042)	--- 6/	.343 (8.71)	.119 (3.02)	.112 (2.84)
-5	.313 (7.95)	.099	.114 (0.052)	--- 6/	.406 (10.31)	.174 (4.42)	.164 (4.17)
-6	.375 (9.53)	.123	.139 (0.063)	--- 6/	.469 (11.91)	.230 (5.84)	.218 (5.54)
-8	.500 (12.70)	.158	.263 (0.119)	.121 (0.055)	.585 (14.86)	.306 (7.77)	.289 (7.34)
-10	.625 (15.88)	.205	.377 (0.171)	.173 (0.078)	.687 (17.45)	.387 (9.83)	.366 (9.30)
-12	.750 (19.05)	.327	.442 (0.200)	.203 (0.092)	.812 (20.62)	.493 (12.52)	.466 (11.84)
-16Z 7/	1.000 (25.40)	.580	.864 (0.392)	.387 (0.175)	1.140 (28.96)	.700 17.78	.661 (16.79)
-20Z 7/	1.250 (31.75)	.746	1.373 (0.623)	.661 (0.300)	1.390 (35.31)	.900 (22.86)	.850 (21.59)
-24Z 7/	1.500 (38.10)	.972	1.599 (0.725)	.948 (0.430)	1.707 (43.36)	1.125 (28.58)	1.063 (27.00)

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ These ball sizes came from SAE-AS1946.
4/ Minimum specified inside diameter shall be verified by passing a spherical ball through the hose assembly.
5/ Swivel nut and cone seat of nipple size- 3/-4 shall mate with the size 3 fitting of SAE-AS4395 or SAE-AS33514 or SAE-AS4375, as specified on the individual slash sheets. The remaining portion of the fitting shall mate with size 4 hose.
6/ Maximum weight for sizes -3/-4 through size 6 does not exist (see 1.2 for class 2).
7/ The letter “Z” is used to indicate that two layers of reinforcement are required for these sizes (see MIL-DTL-32435).

3.5.6 Assembly passage-way inspection. The assembly, with attached fittings, shall be examined for any excessive restriction to the fitting passage by rolling a ball of the applicable diameter in table II, through the assembly. The ball shall pass freely through the assembly. The ball checks the ovality of the bend (tube) and the match-up of the tube and nipple or connection end and the weld/braze drop through (see 4.7.1).

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3.6 Performance.

3.6.1 Operating temperature and pressure. Class 1 shall operate in temperature ranging from -65°F to 450°F (-54°C to 232°C). Class 2 shall operate in temperature ranging from -65°F to 275°F (-54°C to 135°C). The operating pressure of the hose assembly shall not be greater than the applicable value specified in table III.

TABLE III. Performance requirements and hose assembly sample lengths. 1/ 2/

Size	Length of samples for impulse test in.(mm) 3/	Length of samples for all other tests in.(mm) 3/	Maximum operating pressure per square inch psi (MPa)	Proof pressure psi (MPa) 4/ 5/	Minimum room temperature burst pressure psi (MPa)	Minimum high temperature burst pressure psi (MPa)	Minimum inside bend radius inches (mm)
-3/-4	14.0 (355.6)	18.0 (457.2)	1500 (10.34)	3000 (20.68)	12000 (82.74)	7000 (48.26)	2.00 (50.8)
-4	14.0 (355.6)	18.0 (457.2)	1500 (10.34)	3000 (20.68)	12000 (82.74)	7000 (48.26)	2.00 (50.8)
-5	16.0 (406.4)	18.0 (457.2)	1500 (10.34)	3000 (20.68)	10000 (68.95)	6500 (44.82)	2.00 (50.8)
-6	18.0 (457.2)	18.0 (457.2)	1500 (10.34)	3000 (20.68)	9000 (62.05)	6500 (44.82)	4.00 (101.6)
-8	21.0 (533.4)	18.0 (457.2)	1500 (10.34)	3000 (20.68)	8000 (55.16)	6000 (41.37)	4.63 (117.6)
-10	23.5 (596.9)	18.0 (457.2)	1500 (10.34)	3000 (20.68)	7000 (48.26)	5500 (37.92)	5.50 (139.7)
-12	27.5 (698.5)	18.0 (457.2)	1000 (6.89)	2000 (13.80)	5000 (34.47)	3500 (24.13)	6.50 (165.1)
-16Z	18.0 (457.2)	18.0 6/ (457.2)	1250 (8.62)	2500 (17.24)	5000 (34.47)	3500 (24.13)	7.38 (187.4)
-20Z	18.0 (457.2)	18.0 6/ (457.2)	1000 (6.89)	2000 (13.80)	4000 (27.58)	3000 (20.68)	11.00 (279.4)
-24Z	18.0 (457.2)	18.0 6/ (457.2)	1000 (6.89)	2000 (13.80)	4000 (27.58)	3000 (20.68)	14.00 (355.6)

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ The number of samples required for first article testing as specified in 4.5.1.
4/ Assemblies with aluminum flange fittings shall be proof-pressure tested to the values listed under the proof pressure column as a minimum.
5/ Proof-pressure tolerances are +5%, -0%.
6/ Samples for the low temperature flexibility and the vacuum tests shall be 30 inches (762 mm) long for these sizes.

3.6.2 Proof pressure. When subjected to the applicable proof pressure specified in table III and tested in accordance with 4.7.2, the hose assembly shall not leak through the wall of the hose or around the end fittings. The fittings shall not exhibit any visual evidence of damage or permanent deformation.

3.6.3 Leakage. There shall be no leakage through the wall of the hose or around the fittings when the hose assembly is subjected to testing in accordance with 4.7.3.

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3.6.4 Room temperature burst pressure. When subjected to testing in accordance with 4.7.4, the hose assembly shall not leak or burst below the applicable room temperature burst pressure specified in table III. There shall be no leakage around the end fittings, and the end fittings shall not loosen or separate from the hose.

3.6.5 High temperature burst pressure. When subjected to testing in accordance with 4.7.5, the hose assembly shall not leak or burst below the applicable high temperature burst pressure specified in table III. There shall be no leakage around the end fittings, and the end fittings shall not loosen or separate from the hose.

3.6.6 Hose conductivity. Hose assemblies when subjected to conductivity testing in accordance with 4.7.6, using a test potential of 1,000 V dc, hose assembly sizes -3/-4 through 8 shall conduct a direct current of not less than 6 μA. Hose assembly sizes -10 through -24Z shall conduct a direct current of not less than 12 μA.

3.6.7 Impulse. The hose assembly during impulse testing shall not leak, burst, or loosen from the test apparatus and there shall be no evidence of malfunctioning when subjected to 100,000 cycles in accordance with 4.7.7.

3.6.8 Fuel resistance. The hose assembly when subjected to fuel resistance testing in accordance with 4.7.8, the hose assembly shall not leak or show any evidence of deterioration during and at the completion of the test.

3.6.9 Pneumatic effusion. Hose assemblies when subjected to pneumatic effusion testing in accordance with 4.7.9, the measured effusion rate of the hose assembly shall not be greater than the values listed in table IV.

TABLE IV. Pneumatic effusion requirement. 1/

Size	-3/-4	-4	-5	-6	-8	-10	-12	-16Z	-20Z	-24Z
Effusion rate 2/ (cc/ft of hose)	4.0	4.0	5.0	5.0	5.0	5.0	6.0	8.0	8.0	8.0

1/ Max effusion rates

2/ Collected during the last 30 minutes of the test.

3.6.10 Pneumatic surge. Hose assemblies when subjected to pneumatic surge testing in accordance with 4.7.10, the hose assembly shall not leak through the hose wall or around the end fittings. The inner tube of the hose shall not exhibit any evidence of degradation when the filter is examined at the completion of the test.

3.6.11 Stress degradation. This test is intended to verify that the hose inner tube has been sintered and quenched to the proper crystallinity to eliminate stress cracking or creep with subsequent leakage. The effusion rate of the hose assembly after stress degradation shall not be greater than the values listed in table V when tested in accordance with 4.7.11. Following the introduction of hot oil, the hose assembly shall not leak when subjected to the applicable proof pressure test specified in 4.7.2.

TABLE V. Effusion requirement after stress degradation test. 1/

Size	-3/-4	-4	-5	-6	-8	-10	-12	-16Z	-20Z	-24Z
Effusion rate (cc^3/in/min)	8	8	8	8	4	2	2	2	2	2

1/ Max effusion rates.

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3.6.12 Corrosion. The hose assembly when subjected to corrosion testing in accordance with 4.7.12, the hose assembly shall not leak or malfunction below the applicable room and high temperature pressures specified in table III. The fittings shall be in good mechanical working order and shall also be free of corrosion.

3.6.13 Over-tightening torque. There shall not be any leakage when the fitting (flared or flareless only) is subjected to testing in accordance with 4.7.13. After each disassembly, the nut shall swivel freely by hand.

3.6.14 Elongation and contraction. Hose assemblies when subjected to the applicable operating pressure specified in table III and tested in accordance with 4.7.14, the hose assembly shall not change length by more than +.20 or -.30 inch (+5.08 or -7.62 mm) in 10 inches (254 mm) of length.

3.6.15 Volumetric expansion. The volumetric expansion, measured in cc/in of free length, shall not be greater than .028 inch (0.71 mm) for size -4 and .040 inch (1.02 mm) for size -5 when tested in accordance with 4.7.15.

3.6.16 Low temperature flexibility. Hose assemblies when tested in accordance with 4.7.16 and subjected to the applicable bend radius specified in table III, the hose assembly shall not leak or exhibit any visible evidence of permanent deformation or damage.

3.6.17 Vacuum. The hose shall not collapse or show any evidence of defects when subjected to testing in accordance with 4.7.17. The ball shall roll freely through the entire length of the hose, unaided, to indicate that the inside diameter of the hose has not been distorted or reduced.

3.6.18 Pneumatic leakage. Hose assemblies when subjected to pneumatic leakage testing in accordance with 4.7.18, a steady stream of bubbles shall not be seen from any area of the hose assembly.

3.6.19 Fire resistance. When fire resistance or fireproofing is required (see 6.2), the hose assembly shall not rupture or leak when subjected to testing in accordance with 4.7.19.

3.6.20 Cleanliness. Prior to shipping, the end fittings of the hose assembly shall be capped or plugged to prevent entrance of moisture and foreign matter. The caps or plugs shall be securely attached and shall withstand normal strains, jarring, vibrations encountered during shipping, storage and handling. The interior surface of the hose assembly shall be free from oil, grease, dirt, moisture, cleaning solvents and foreign materials. All hose assemblies shall be cleaned in accordance with SAE-AS611, class O. During conformance inspection, hose assemblies with uncovered ends shall be rejected and considered as a failure. The interior of the assembly shall not contain any debris or foreign materials when examined in accordance with 4.7.20.

3.7 Identification of product.

3.7.1 Fittings. The fittings manufacturer's name, or trademark shall be permanently marked on all end fittings.

3.7.2 Hose assemblies. Marking shall be on permanent band or bands securely attached on the hose. Bands shall not be wider than 1 inch (2.54 mm) and shall not impair the flexibility or the performance of the hose. The bands shall be designed as to remain tight on the hose to prevent relative movement and resultant chafing. Unless other wise specified, the marking on the band shall include the following information:

a. Assembly manufactures Commercial and Government Entity (CAGE) code
b. Part or Identifying Number (PIN) as specified in the applicable specification sheet.
c. Operating pressure (3000 psi) as applicable to table III.
d. Operating temperature (450ºF or 275ºF) as applicable see paragraph 1.2.
e. Pressure test symbol (PT).
f. Date of hose assembly manufacture expressed as month and year (MM/YY).
g. Hose manufacturers CAGE if different than the hose assembly manufacture.

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3.8 Interchangeability. All hose assemblies having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.9 Workmanship. All surfaces shall be free from burrs. All sealing surfaces shall be smooth, except that annular tool marks shall not be greater than 100 microinches.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5).
b. Conformance inspection (see 4.6).

4.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained or identified by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with ISO 17025 and NCSL Z540.3 as applicable.

4.3 Inspection conditions. Unless otherwise specified, all required inspections shall be performed in accordance with the test conditions specified in 4.7.

4.4 Responsibility for compliance. All items shall meet all requirements of sections 3, 4, and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material. The supplier's inspection and test records shall clearly demonstrate the products conformance to technical specifications.

4.4.1 Lot records. Manufacturers shall keep lot records for 3 years minimum. Manufacturers shall monitor for compliance to the prescribed procedures, and observe that satisfactory manufacturing conditions and records on lots are maintained for these hose assemblies. The records, including as a minimum, an attributes summary of all quality conformance inspections conducted on each lot, shall be available to review by customers at all times.

4.5 First article inspection. All hose assembly samples shall be subjected to first article testing in accordance with table VI and in the sequence specified in table VII. First article inspection shall consist of the examinations and tests specified in performed on samples which are representative of the production item, after award of the contract, to determine that the production item meets the requirements of this specification.

4.5.1 Samples for first article. Test samples shall consist of the number of samples and lengths specified in table III. If field-attachable fittings are used, the fittings shall comply with MIL-DTL-27272 and be used with MIL-DTL-32435 hose. Where permanent end fittings are used and more than one source of hose or hose liner are used, first article tests shall be conducted on test samples constructed using hose liner from each source.

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4.5.1.1 First article samples. First article samples shall be representative of the product proposed to be furnished to this specification. Samples consisting of 16 hose assemblies of the same size and class, with lengths as specified in table III or in the applicable test method.

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a. Reusable fittings. All assemblies with reusable fittings shall be assembled in accordance with the instructions specified in MIL-DTL-27272. When permanent fittings are used in the hose assemblies, two separate end fittings of the same type of interface, size, and class shall also be submitted for first article testing (see table VI).

b. Permanent fittings. Hose assemblies with permanent fittings shall be first article tested with fittings from a specific manufacturer(s) and bulk hose from a specific manufacturer(s).

c. Both reusable and permanent fittings. For assemblies with either permanent or reusable fittings, the assemblies shall be first article tested with fittings from a specific manufacturer and hose from a specific manufacturer. If multiple hose sources and/or multiple fitting sources are involved, each combination must be first article inspected.

4.5.2 Waivers or deviations to specification requirements. All waivers or deviations to specification requirements shall be coordinated through the preparing activity; DLA Land and Maritime, Attn: VAI, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Fluidflow@dla.mil.

4.5.3 First article information. Upon completion of first article inspection, the Government activity responsible for conducting the inspection program (see 6.2), shall report the results of the inspection, with appropriate recommendation, to the contracting officer. Approval of the first article samples or the waiving of first article inspection does not preclude the requirements for performing conformance inspection.

4.5.3.1 Disposition of samples. First article samples shall be furnished to the Government as directed by the contracting officer (see 6.2).

4.5.3.2 Failures. All samples shall meet all of the contract requirements. Failure of a sample unit to pass any test shall be cause for rejection of the entire lot and refusal to grant first article approval.

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TABLE VI. First article inspection requirements.

Requirement	Requirement paragraph	Test method paragraph
Examination of product	3.5	4.7.1
Proof pressure	3.6.2	4.7.2
Leakage	3.6.3	4.7.3
Room temperature burst pressure	3.6.4	4.7.4
High temperature burst pressure	3.6.5	4.7.5
Hose assembly conductivity	3.6.6	4.7.6
Impulse	3.6.7	4.7.7
Fuel resistance	3.6.8	4.7.8
Pneumatic effusion	3.6.9	4.7.9
Pneumatic surge	3.6.10	4.7.10
Stress degradation	3.6.11	4.7.11
Corrosion	3.6.12	4.7.12
Over-tightening torque	3.6.13	4.7.13
Elongation & contraction	3.6.14	4.7.14
Volumetric expansion	3.6.15	4.7.15
Low temperature flexibility	3.6.16	4.7.16
Vacuum	3.6.17	4.7.17
Pneumatic leakage	3.6.18	4.7.18
Fire resistance or fire proof	3.6.19	4.7.19

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TABLE VII. First article inspection sequence.

Required first article test	Required test paragraph	Sample number											
		Fittings 1/ (permanent)	Assemblies (see table III).										
			1 2/	2	3	4	5	6 3/	7	8 2/	9	10 2/	11-16 3/
Examination of product	4.7.1	X	X	X	X	X	X	X	X	X	X	X	X
Proof pressure	4.7.2		X	X	X	X	X	X	X	X	X	X	X
Stress degradation	4.7.11								X	X			
Corrosion	4.7.12				X	X							
Over-tightening torque 4/	4.7.13	X											
Elongation & contraction	4.7.14			X								X	
Volumetric expansion	4.7.15			X							X	X	
Pneumatic leakage	4.7.18						X	X					
Hose conductivity	4.7.6										X		
Impulse	4.7.7												X
Fuel resistance	4.7.8		X	X									
Pneumatic effusion	4.7.9						X	X					
Pneumatic surge	4.7.10						X	X					
Low temp flexibility	4.7.16		X							X		X	
Vacuum	4.7.17		X							X		X	
Proof pressure	4.7.2						X	X					
Leakage	4.7.3						X	X					
Room temp burst pressure	4.7.4				X		X						
High temp burst pressure	4.7.5					X		X					

1/ Two fittings of each type of interface are required only when first article of assemblies using permanent fittings is desired (see 4.5.1.1(b)).

2/ These samples shall be 30 inches in length for sizes -16Z, -20Z and -24Z.

3/ These samples shall have a 90°elbow fitting on one end and a straight fitting on the other end. If first article is desired for both bent tube and forged elbow, then half of the samples shall use the other type. In this case, sample 6 may have either a bent tube or a forged configuration.

4/ Not applicable to flange fittings.

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4.6. Conformance inspection. For manufacturers that have successfully passed first article inspections and are continuously producing hose assemblies to this specification, on going inspections shall consist of individual inspections (see table VIII). If first article is waived due to prior successful first article inspection, the individual inspections and sampling and periodic inspections shall be the manufactures in house inspection procedures.

4.6.1 Individual tests. Inspection of the product for delivery; shall consist of subjecting each hose assembly to the individual tests specified in table VIII. Any item failing to meet the requirements of the individual inspections shall be immediately removed from the lot.

TABLE VIII. Individual inspection.

Requirement	Requirement paragraph	Test method paragraph
Examination of product	3.5	4.7.1
Proof pressure	3.6.2	4.7.2

4.6.2 Nonconformance of individual inspection. If one or more defects are identified, then the entire production lot shall be screened for that defect and all defects shall be removed. A second inspection sample shall then be selected from the different lot and the individual inspections shall be performed again. If one or more defects are identified from the second inspection lot, then the entire production lot shall be rejected and not supplied to this specification.

4.7 Test methods. Test methods used shall be as specified in table VI. For first article testing, the sequence of tests performed shall be in accordance with table VII. Room temperature shall be defined as 60 to 100°F (16°C to 38°C) unless otherwise specified. Class 2 hose assemblies shall be tested at 275°F (135°C) whenever a higher temperature is specified.

4.7.1 Examination of product. Each hose assembly shall be visually and physically examined for conformance to the configuration specified in the applicable specification sheet and the following requirements:

a. Materials (see 3.3).
b. Design and construction (see 3.5).
c. Dimensions and weights (see 3.5.4).
d. Cleanliness (see 3.6.20).
e. Product identification (see 3.7).
f. Interchangeability (see 3.8).
g. Workmanship (see 3.9).
h. Assembly passage-way inspection (see 3.5.6)

4.7.2 Proof pressure. Prior to testing, each hose assembly shall be examined to ensure that it is properly assembled. Samples shall then be subjected to testing in accordance with SAE-AS2078 using the applicable proof pressure as specified in table III. However, hose assemblies with aluminum flange fittings (class 2) shall be proof tested using the applicable operating pressure listed in table III. Test fluid shall be water; however, fluid in accordance with MIL-PRF-5606, MIL-PRF-83282, or MIL-PRF-87257 may be used during first article testing. Conformance shall be as specified in 3.6.2.

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4.7.3 Leakage. Prior to testing, the exterior of the samples shall be cleaned free of oil and of any residues. Samples shall be subjected to testing in accordance with SAE-AS2078. Test fluid shall be water or fluid in accordance with MIL-PRF-5606, MIL-PRF-83282, or MIL-PRF-87257. A white paper towel, wrapped tightly around the entire length of the sample, shall be used to check for leakage. Conformance shall be as specified in 3.6.3.

4.7.4 Room temperature burst pressure. Samples shall be subjected to testing in accordance with SAE-AS2078 using the applicable room temperature burst pressure specified in table III. Test fluid shall be water or fluid in accordance with MIL-PRF-5606, MIL-PRF-83282, or MIL-PRF-87257. Samples shall be continuously observed throughout the duration of the test. The type of failure and the pressure at which failure occurred shall be recorded. Conformance shall be as specified in 3.6.4.

4.7.5 High temperature burst pressure. Samples shall be subjected to testing in accordance with SAE-AS2078 using the applicable high temperature pressure specified in table III. Test fluid shall conform to MIL-PRF-7808, MIL-PRF-83282, or MIL-PRF-87257. Samples shall be continuously observed for the duration of the test. The type of failure and the pressure at which failure occurred shall be recorded. Conformance shall be as specified in 3.6.5.

4.7.6 Hose conductivity. The test sample shall be a 13-inch (330 mm) length of hose with a fitting attached at only one end. Testing shall be in accordance with the conductivity test for hose assembly specified in SAE-AS2078. Conformance shall be as specified in 3.6.6.

4.7.7 Impulse (see 3.6.7). Hose assemblies when subjected to impulse testing shall meet the requirements of 3.6.7, the following details shall apply:

a. All test specimens, of length as specified in table III, shall be subjected to the proof pressure test (see 4.5.2) prior to impulse testing in accordance with SAE-AS2078.
b. For first article testing, 2 of the 6 samples shall be unaged while 2 samples shall be aged in air at 400°F ±10°F (204°C ±5.5°C) for 168 hours. The remaining 2 samples shall be aged by immersion in fluid in accordance with MIL-PRF-7808 at 400°F ±10°F (204°C ±5.5°C) for 168 hours.
    (1) The assemblies shall then be subjected at room temperature to the proof pressure specified in table III for a minimum of 5 minutes.
    (2) The hose assemblies shall then be pressurized to operating pressure and while maintaining this pressure at room temperature, the hose assemblies shall be immersed in a 3.5% +0.1% U.S.P. Grade NaCI solution by weight for 8 to 10 minutes, then allow to air dry for the remainder of 1 hour. This sequence of immersion and air drying shall be repeated no less than 50 times.
c. Samples sizes 12 and smaller shall be connected to the rigid supports of the test apparatus and bent to the applicable radius specified in table III. Samples sizes -16Z and larger shall be installed straight; one end may be left free.
d. The peak pressure used for samples shall be 100% of operating pressure specified in table III.

4.7.8 Fuel resistance (see 3.6.8). Samples shall be subjected to the fuel resistance test in accordance with SAE-AS2078 shall meet the requirements of 3.6.8.

4.7.9 Pneumatic effusion (see 3.6.9). Samples shall be subjected to the pneumatic effusion test in accordance with SAE-AS2078 shall meet the requirements of 3.6.9.

4.7.10 Pneumatic surge (see 3.6.10). Samples shall be subjected to the pneumatic surge test in accordance with SAE-AS2078 shall meet the requirements of 3.6.10.

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4.7.11 Stress degradation (see 3.6.11). Hose assemblies when stress degradation tested shall meet the requirements of 3.6.11, the following details shall apply:

a. Two hose assemblies of each size shall be subjected to this test. The hose assemblies shall be filled with high temperature test fluid in accordance with MIL-PRF-7808 or MIL-PRF-83282.
b. The hose assemblies shall then be placed in an oven which shall be maintained at a temperature of 450°F ±10°F (232°C ±5.5°C). Precautions shall be taken to assure that the hose assemblies do not come in contact with part of the oven that are at a higher temperature. A pressure equal to the rated operating pressure specified in table III shall be applied to the hose assemblies.
c. After a minimum of 20 hours at 450°F ±10°F (232°C ±5.5°C), the pressure shall be gradually released and the assemblies shall be removed from the oven, drained and cooled to room temperature. The assemblies shall then be flushed with a quantity of new test fluid, equivalent in volume to at least twice the sample volume and drained.
d. The hose assemblies shall then be filled with MIL-PRF-87257 fluid. A pressure equal to the rated operating pressure specified in table III shall be applied and held for a minimum of 2 hours at room temperature.
e. The assemblies shall then be emptied and filled with oil or hydraulic fluid as specified in 4.7.11 (a). The tests specified in 4.7.11 (b), (c), and (d) shall be repeated.
f. The hose assemblies shall then be filled with ASTM reference fluid B (isooctane, 70%; toluene, 30%) in accordance with ASTM D471 and individually capped. While at room temperature, the assemblies shall be bent around a mandrel having a radius equal to the minimum bend radius as specified in table III. The assemblies shall be bent around the mandrel and straightened for 20 cycles. The assemblies shall be held by the fitting while the bending is being performed. The tests specified in 4.7.11 (a), (b), (c), and (d) shall be conducted for the third time.
g. Within 4 hours after the final 2 hours pressurization period with ASTM reference fluid B, the assemblies shall be drained and flushed with fluid in accordance with MIL-PRF-680 and placed in an oven for 1 hour. The temperature of the oven shall be maintained at 160°F ±10°F (71°C ±5.5°C).
h. Within 8 hours after completion of the drying process, the hose assemblies shall be removed from the oven, cooled to room temperature, and then subjected to an air-under-water test. To conduct this test, the hose assemblies shall be installed in an apparatus similar to that shown on figure 2.

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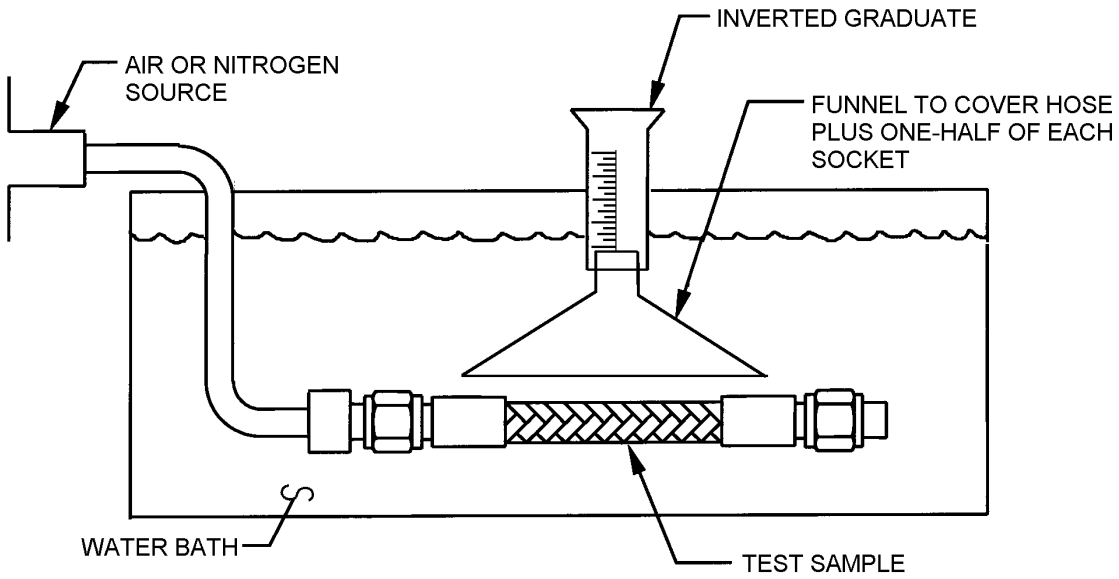


FIGURE 2. Test setup for pneumatic effusion tests.

i. This test setup with the hose assembly installed shall be immersed in water containing no wetting agent. A pressure equivalent to the rated operating pressure, as specified in table III, shall be applied and the pressure shall be held for a minimum of 15 minutes to allow any entrapped air in the hose assembly to escape.

j. The pressure shall be held for an additional 5 minute period. During this time effused gas shall be collected from the test sample which includes the juncture of the hose to the fitting, but not including the fitting nut. If after the 5 minute period of pressurization, the average rate of effusion through the hose assembly exceeds the values listed in table V, it shall be cause for rejection.

k. At the completion of the tests specified in 4.7.11 (b) through (j), the hose assemblies shall be filled with oil and placed in a cold chamber for 8 hours while maintained at -67°F ± 2°F (-55°C ±1.1°C)

l. After the 8 hour cold soak, the assemblies shall be subjected to a pressure equal to the operating pressure specified in table III. The pressure shall be held for a minimum of 5 minutes and then released. This shall be repeated for a total of 10 times with a minimum of 5 minutes between each pressure application and with the samples still in the -67°F ±2°F (-55°C ±1.1°C) cold chamber.

m. At the end of this time oil at a temperature of 450°F ±10°F (232°F ±5.5°C) shall be circulated through the hose assemblies.

n. Within 15 seconds after introduction of the hot oil, the pressure shall be increased to the rated proof pressure specified in table III and held for 2 minutes. There shall be no evidence of leakage from the hose.

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4.7.12 Corrosion (see 3.6.12). Hose assemblies when subjected to corrosion testing shall meet the requirements of 3.6.12, the following details shall apply:

a. Samples, mounted in a vertical position and subjected to the applicable operating pressure (see table III), shall be immersed in a 2.5% solution of sodium chloride for five minutes.
b. They shall then be air-dried at 140°F ±10°F (60°C ±5.5°C) for 25 minutes.
c. This cycling shall continue for at least 172 hours with the specified pressure maintained on the hose.
d. Following the cycling, half of the samples shall be subjected to the room temperature burst pressure test (see 4.7.4) and the other half shall be subjected to the high temperature burst pressure test (see 4.7.5).

4.7.13 Over-tightening torque (see 3.6.13). Hose assembles when subjected to over-tightening torque in accordance with SAE-ARP908 shall meet the requirements of 3.6.13. The following exception shall apply: The number of cycles used shall be 15.

4.7.14 Elongation and contraction (see 3.6.14). Hose assembles when subjected to elongation and contraction testing in accordance with SAE-AS2078 shall meet the requirements of 3.6.14. The test fluid used shall be water or fluid in accordance with MIL-PRF-5606, MIL-PRF-83282, or MIL-PRF-87257.

4.7.15 Volumetric expansion sizes 4 and 5 only (see 3.6.15). Hose assembles when subjected to volumetric expansion be in accordance with SAE-AS2078 shall meet the requirements of 3.6.15. The following exception shall apply: the operating pressure used shall be 1000 psi (7 Mpa).

4.7.16 Low temperature flexibility (see 3.6.16). Hose assembles when subjected to low temperature flexibility testing in accordance with SAE-AS2078 shall meet the requirements of 3.6.16, the following exception shall apply: Sample lengths shall be as specified in table III. After low temperature flexibility testing these samples shall be vacuum tested, see 4.7.17.

4.7.17 Vacuum (see 3.6.17). Hose assembles when subjected to vacuum testing in accordance with SAE-AS2078 shall meet the requirements of 3.6.17. The following details shall apply:

a. Samples that were tested for low temperature flexibility (see 4.7.16) shall then be subjected to vacuum testing in accordance with SAE-AS2078.
b. The applicable negative pressure used shall be as specified in table VIII and ball diameters shall be as specified in SAE-AS1946.

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TABLE VIII. Vacuum test conditions. 1/ 2/

Size	Nominal inside diameter inch (mm)	Negative pressure inHg (kPa)
-3/-4	0.188 (4.78)	28 (94.8)
-4	0.188 (4.78)	28 (94.8)
-5	0.250 (6.35)	28 (94.8)
-6	0.313 (7.95)	28 (94.8)
-8	0.406 (10.31)	28 (94.8)
-10	0.500 (12.70)	28 (94.8)
-12	0.625 (15.88)	20 (67.7)
-16Z	0.875 (22.23)	14 (47.4)
-20Z	1.125 (28.58)	10 (33.9)
-24Z	1.375 (34.93)	8 (27.1)

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

4.7.18 Pneumatic leakage (see 3.6.18). Hose assembles when subjected to pneumatic leakage in accordance with SAE-AS2078 shall meet the requirements of 3.6.18.

4.7.19 Fire resistance (see 3.6.19). Hose assembles when subjected to fire resistance testing in accordance with SAE-AS1055 type IIa, class A, shall meet the requirements of 3.6.19.

4.7.20 Cleanliness (see 3.6.20). Both ends of the hose assembly shall be visually inspected to determine if caps or plugs are installed at the fittings. For hose assembly with straight fittings at both ends, remove the caps or plugs and place a light source at one end of the hose assembly. The interior of the hose shall be visually examined, without magnification, from the opposite end of the light source. Conformance shall be as specified in 3.6.20.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

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6.1 Intended use. The hose assembly covered by this specification is military unique to meet the requirements specified herein and in the applicable specification sheets. The hose in accordance with MIL-DTL-32435 for field attachable fitting assemblies should be mated with either reusable fittings qualified in accordance with MIL-DTL-27272 or with permanent fittings to meet the requirements specified herein and in the applicable specification sheet. It is recommended that this hose assembly should not be used in pneumatic storage system applications and in any applications that exceed the operating temperature range and pressure specified herein. Hose assembly sizes -20Z and -24Z should not be used in systems where the peak pressures are greater than 1000 psi (7MPa). Hose assembly size -3/-4 is not intended for use in aircraft hydraulic systems.

6.1.1 Military unique rationale. These hose assemblies are military unique because they are intended for use in high-temperature fuel, lubricating oil, water-alcohol, chemical-fluid, hydraulic and pneumatic systems that allow some gaseous effusion through the hose wall. However, hose assembly used in hydraulic and pneumatic systems should not exceed the temperature limit of 400°F (204°C). This hose assembly is a hydraulic or pneumatic conductor for a fluid at the operating pressure, specified in table III, even under extreme temperature conditions ranging from -65°F to 450°F (-54°C to 232°C) (275°F (135°C) for class 2).

6.1.2 Fire resistance. When fire proofing or fire resistance is desired, it is critical that the hose assembly meets the requirements specified in 3.6.19 when tested in accordance with 4.7.19.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. PIN (see 1.2)
c. First article (see 4.5) or whether first article inspection is waived (see 6.3.1).
d. Lot records if required (see 4.4.1).
e. Name and address of the first article inspection test facility to which first article samples, if required, are to be forwarded (see 4.5) and the name and address of the Government activity responsible for conducting the first article inspection program (see 6.3).
f. Packaging requirements (see 5.1).

6.3 First article. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first article samples. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 Defense Logistics Agency (DLA) waiver of first article test. A waiver of a first article testing will only be considered by DLA when the contractor has delivered the same item within the last three years, has no unfavorable quality history, has not changed processes, or changed any subcontractors. DLA will not accept first article testing results outside the stated requirements.

6.4 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

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6.5 Subject term (key word) listing.

2,2,4-Trimethylpentane (Isooctane)
Fire resistance
Fuel resistance
Hydraulic systems
Lubricating oil
Passivated
Pneumatic systems
Toluene
Water-alcohol

6.6 Guidance on use of alternative parts with less hazardous or nonhazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit and function requirements of their application.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2012-018)

Review activities:
Army - AR, AT, EA, MI
Navy - MC, SA, SH
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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